

Project for the "Local Volatility Model"

1) In order to optimize the calibrator.m function, we modified the solve_dupire.m function by changing line 40 such as

```
C(:,i+1) = (I+0.5*dt*Ap)\( (I-0.5*dt*A)*C(:,i) );
```

In this way the function memorizes every time step, giving us an approximate information about the call price with respect to the value of the call at the previous expiry.

Then inside the function model_volatility.m in order to have just one Dupire equation solved we took out from the for cycle the following equation:

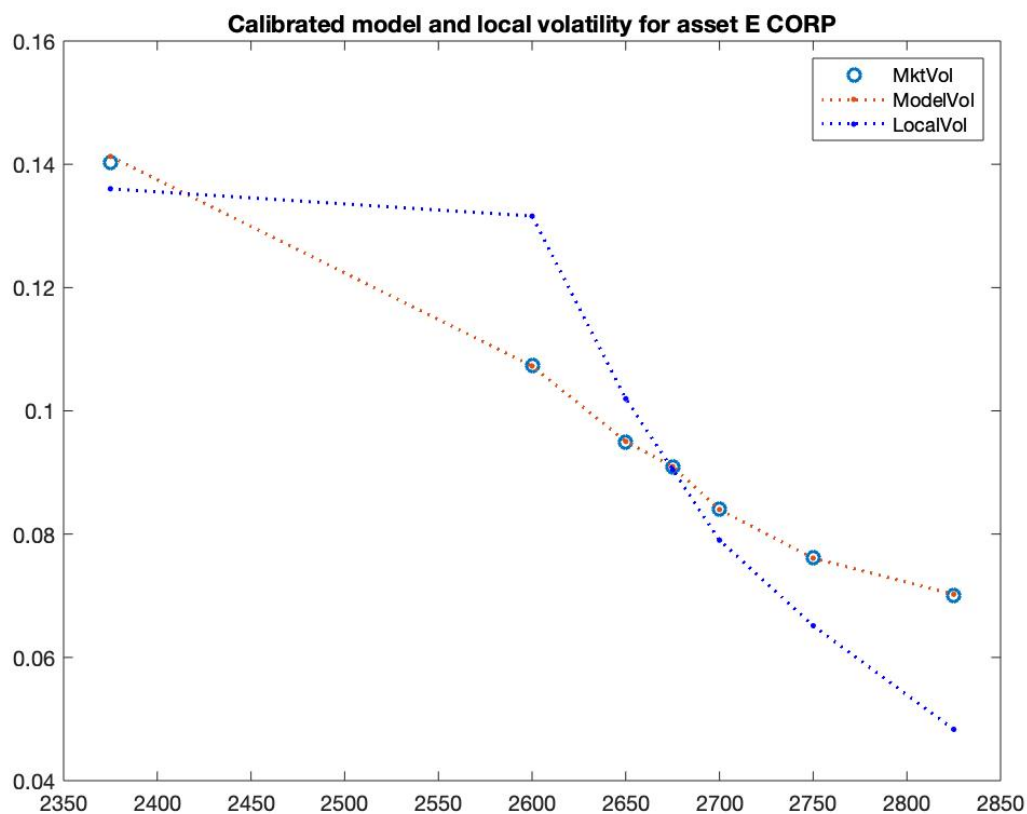
```
expiry = T(end);
```

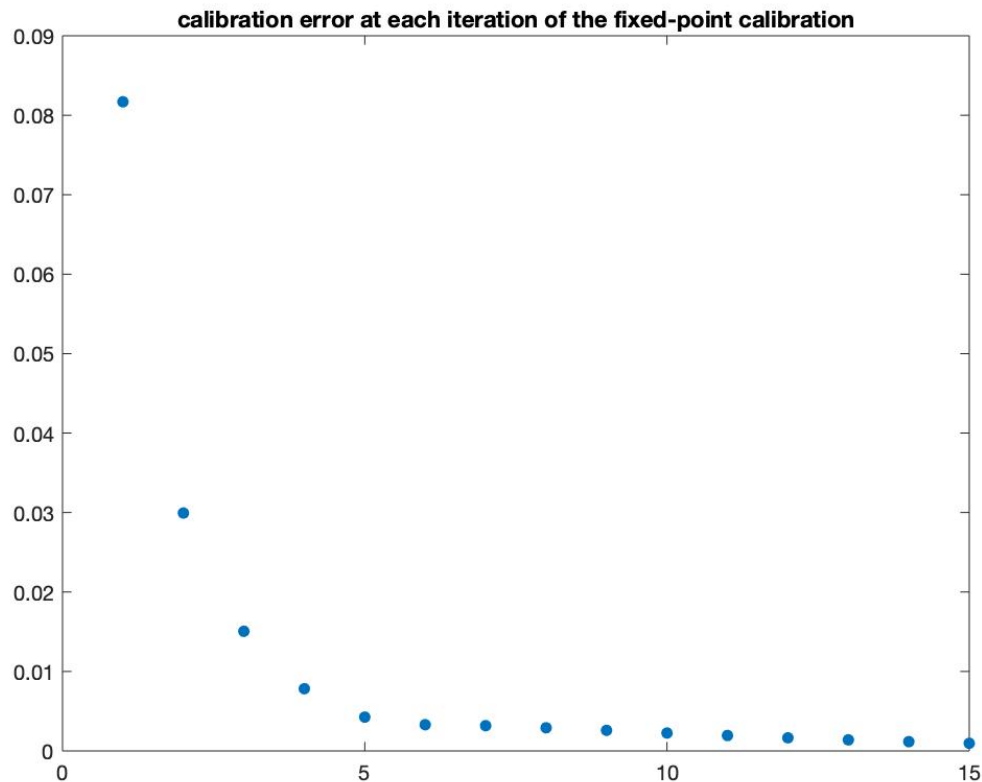
```
[ k, t, C ] = solve_dupire_2( T, K_norm, V, expiry, Lt, Lh, K_min, K_max, Scheme);
```

More precisely, we use the function blsimpv to calculate the implied volatility of the current iteration.

Furthermore we noticed that with a small Lt (time-steps inside the solve_dupire.m) the calibration was not converging because the interpolated results for the prices were not close enough, so we decided to increase it to 120, in order to obtain a stable calibration.

2.1) Plot local volatility function vs market implied volatility





2.2)

lv_price1 =295.3231

lv_price2 =3.4925

lv_impl_vol1 = 0.1572

lv_impl_vol2 =0.0813

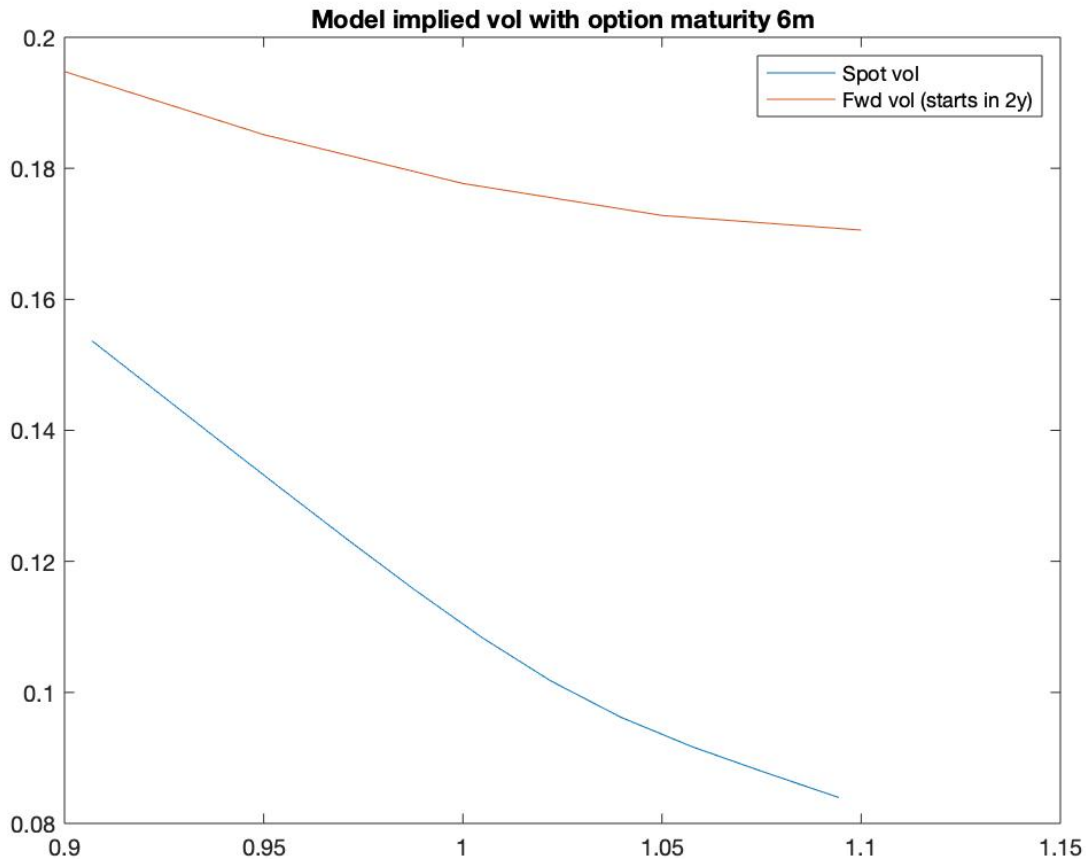
2.3)

model_impl_vol = [0.1950 0.1708]

2.4) skew_spot= -0.3718

Skew_forward=-0.1210

The first thing we notice is that the skew of the forward smile is clearly bigger than the skew of the spot smile, this result is compatible with the literature of the local volatility model. Indeed forward smile flattening is a typical behavior of the LV model and the forward smile flattens as T_1 increases, while this is not a correct representation of the market, where the spot and forward smile should be similar, our data is calculated using the LV model and reflect the same errors of the model.



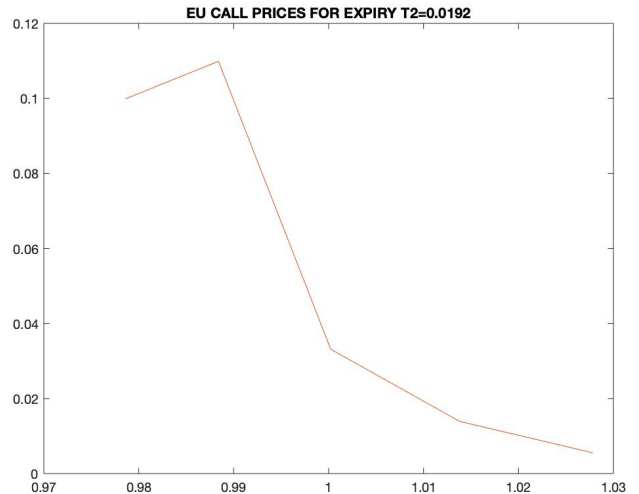
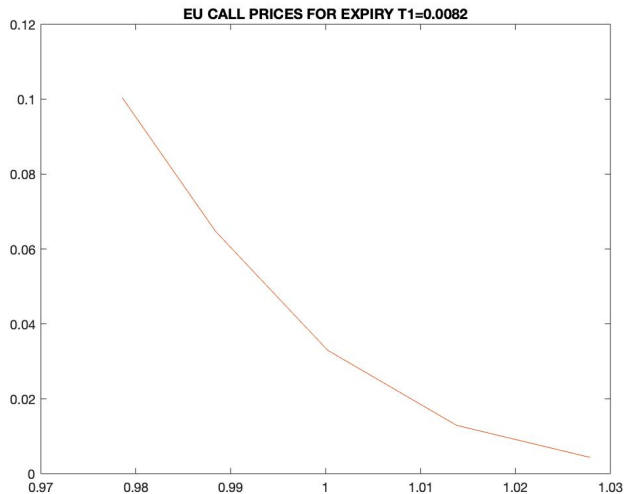
2.5) No it isn't. As a matter of fact forward starting options prices depend on the forward density as well, and each model generates its own forward densities. Derivatives which strongly depends on forward transition densities will typically be mispriced under the lv model.

3) In order to avoid arbitrage in a smooth price function we must have these constraints satisfied:

$$(1 - k)^+ \leq c_0(T, k) \leq 1 \quad -1 \leq \frac{\partial}{\partial k} c_0(T, k) \leq 0 \quad \frac{\partial^2}{\partial^2 k} c_0(T, k) > 0 \quad \frac{\partial}{\partial T} c_0(T, k) \geq 0$$

After plotting the call prices with respect to the two different expiries, respectively T1 and T2, it can be noticed that the second graph is convex, therefore the third constraint is not satisfied, which implies the presence of arbitrage in the market data.

The calibration proceeds to fail because the function calibration.m is not written to accept market data with arbitrage.



4.1)K_eurusd =

130.3395	130.1020	128.8683	127.4776	124.3185	121.8320	119.7897
132.6002	132.4532	131.7354	131.1297	129.5720	128.4054	127.4230
134.7139	134.7537	134.8189	134.8940	135.1111	135.3187	135.5150
136.6468	137.1795	137.7939	138.6073	140.5766	142.2180	143.6756
138.3749	139.3919	140.6851	142.0872	145.8210	148.8043	151.5057

Calibration function outputs: V, ModelVol and MaxErr.

% V is the calibrated local volatility matrix

V =

0.0993	0.0420	0.0736	0.1068	0.1033	0.1101	0.1119
0.0816	0.0323	0.0796	0.0885	0.0938	0.0962	0.0985
0.0648	0.0614	0.0694	0.0859	0.0853	0.0870	0.0898
0.0629	0.0786	0.0609	0.0839	0.0834	0.0870	0.0902
0.0622	0.0740	0.0761	0.0840	0.0893	0.0883	0.0913

% ModelVol is the model implied volatilities at the normalized strikes

ModelVol =

0.0792	0.0699	0.0722	0.0787	0.0864	0.0907	0.0934
0.0722	0.0656	0.0708	0.0752	0.0829	0.0864	0.0889
0.0671	0.0662	0.0682	0.0732	0.0797	0.0829	0.0854
0.0655	0.0680	0.0667	0.0722	0.0784	0.0818	0.0842
0.0648	0.0682	0.0683	0.0723	0.0789	0.0818	0.0841

% MaxErr is the vector of calibration error at each iteration

MaxErr =

0.0107	0.0047	0.0027	0.0016	0.0008
--------	--------	--------	--------	--------

4.2)

lv_vanilla_price = 6.5704

err_vanilla = 0.0051

lv_digital_price = 0.7509

err_digital = 0.0013

4.3)

vanilla_price_black = 6.5856

err_vanilla_black = 0.0052

digital_price_black = 0.7293

err_digital_black = 0.0013

4.4) We can observe that the confidence intervals of the priced call options are very close, while this does not happen when taking into consideration the digital options. The reason is associated with the limitation of the Black model since it cannot reproduce exactly all market prices.

The Black model is a quotation procedure which is calibrated by using discounted factors related to forward and plain vanilla options. This implies that it does not perform precise simulations for other kinds of derivatives assets such as digital options.